LESSON 7.9

Reflect Points Across Axes

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can reflect points across the x-axis and y-axis on the coordinate plane.

WHY THIS MATTERS Captain Vega discovers that her treasure map has a magic mirror trick! When she folds the map along the x-axis, some clues match up perfectly with clues on the other side. For example, a marker at (2, 4) has a matching marker at (2, -4) when reflected over the x-axis. Another pair reflects over the y-axis: (5, 3) and (-5, 3). How does reflection change the coordinates?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Reflection *Spanish: Reflexión*	A flipped, mirror image across a line.	(3, 2) reflected over the y-axis becomes (-3, 2) — x changes sign, y stays
x-axis *Spanish: Eje x*	The line that goes across the grid. Flipping over it changes the y sign.	(4, 3) → (4, -3): the y flips from +3 to -3, like folding the paper along the horizontal line
y-axis *Spanish: Eje y*	The line that goes up the grid. Flipping over it changes the x sign.	(4, 3) → (-4, 3): the x flips from +4 to -4, like folding the paper along the vertical line
Symmetry *Spanish: Simetría*	When a shape looks the same on both sides of a line.	A butterfly's wings are symmetric — fold it down the middle and both sides match
Integer *Spanish: Número entero*	Whole numbers and their opposites, like -2, -1, 0, 1, 2.	..., -3, -2, -1, 0, 1, 2, 3, ...
Perpendicular *Spanish: Perpendicular*	Two lines that meet to make a square corner (90 degrees).	The dashed height meets the base at a square corner — the small 90° box is what makes it perpendicular, not slanted.

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

I will reflect (3, 2) over the y-axis.

1. The y-axis is the vertical line, so the point flips left or right. That changes the x-coordinate.
2. The x changes sign: 3 becomes -3. The y stays the same: 2 stays 2.
3. So the reflection of (3, 2) over the y-axis is (-3, 2).

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Now let's reflect (4, -5) over the x-axis. Which coordinate changes over the x-axis? The y-coordinate.

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- When I reflect over the x-axis, the _____ -coordinate changes sign and the _____ -coordinate stays the same. When I reflect over the y-axis, the _____ -coordinate changes sign and the _____ -coordinate stays the same.

Word bank Reflection | x-axis | y-axis | Symmetry | Integer | Perpendicular | reflect | opposite | sign | change | same | mirror

WATCH OUT!

A common mistake

A common mistake in Reflect Points Across Axes is flipping the sign of the wrong coordinate

LESSON 7.9

Reflect Points Across Axes

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can reflect points across the x-axis and y-axis on the coordinate plane.

WHY THIS MATTERS Captain Vega discovers that her treasure map has a magic mirror trick! When she folds the map along the x-axis, some clues match up perfectly with clues on the other side. For example, a marker at (2, 4) has a matching marker at (2, -4) when reflected over the x-axis. Another pair reflects over the y-axis: (5, 3) and (-5, 3). How does reflection change the coordinates?

PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Match each point and axis to its reflection.

_____ (2, 5) over y-axis	A (-2, 5)
_____ (-3, 1) over y-axis	B (3, 1)
_____ (4, -6) over x-axis	C (4, 6)
_____ (-1, -4) over x-axis	D (-1, 4)
_____ (5, 2) over x-axis	E (5, -2)
_____ (-6, 3) over y-axis	F (6, 3)

HINT Read the axis on each card: the y-axis flips the x sign; the x-axis flips the y sign. Only ONE number changes.

2 **Write the name of the correct group on each line.**

Put the steps in order for reflecting (5, -3) over the y-axis.

Groups:

_____ Start with the point (5, -3)

_____ Identify the axis of reflection: y-axis

_____ The y-axis is vertical, so the x-coordinate changes sign

_____ Change x from 5 to -5, keep y as -3

_____ The reflection is (-5, -3)

HINT The steps tell a story: start with the point, identify the axis, apply the rule, then state the result.

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

What is the reflection of (4, -3) over the x-axis?

(A) (-4, -3)

(B) (-4, 3)

(C) (3, -4)

(D) (4, 3)

HINT Reflecting over the x-axis flips the point UP/DOWN — the left/right position doesn't move.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

What is the reflection of (-2, 6) over the y-axis?

(A) (-2, -6)

(B) (2, 6)

(C) (2, -6)

(D) (6, -2)

HINT Reflecting over the y-axis flips LEFT/RIGHT — the height stays the same.

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Which point is on the y-axis?

(A) (-2, 4)

(B) (0, 5)

(C) (3, 3)

(D) (5, 0)

HINT Points on the y-axis have gone 0 units left or right — check the FIRST coordinate.

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 **Find the mistake. Explain it, then show the correct work.**

Spot the Common Mistake

1. **Problem** Reflect the point (6, 4) over the y-axis.
2. **Attempt** The y-axis is vertical, so I will change the y-coordinate: 4 becomes -4. The answer is (6, -4).
3. **Check** The point only moved up and down, but a fold over the vertical y-axis should move it left and right.

Which step has the mistake? Explain what went wrong, then show the correct work.

> **HINT** The y-axis is the vertical line. Folding over it moves a point left or right, so the x-coordinate is what changes.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

What pattern do you notice about the coordinates when reflecting over the x-axis versus the y-axis?

SENTENCE STARTER When I reflect over the x-axis, the ___ -coordinate changes sign and the ___ -coordinate stays the same. When I reflect over the y-axis, the ___ -coordinate changes sign and the ___ -coordinate stays the same.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 7.9

Reflect Points Across Axes

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can reflect points across the x-axis and y-axis on the coordinate plane.

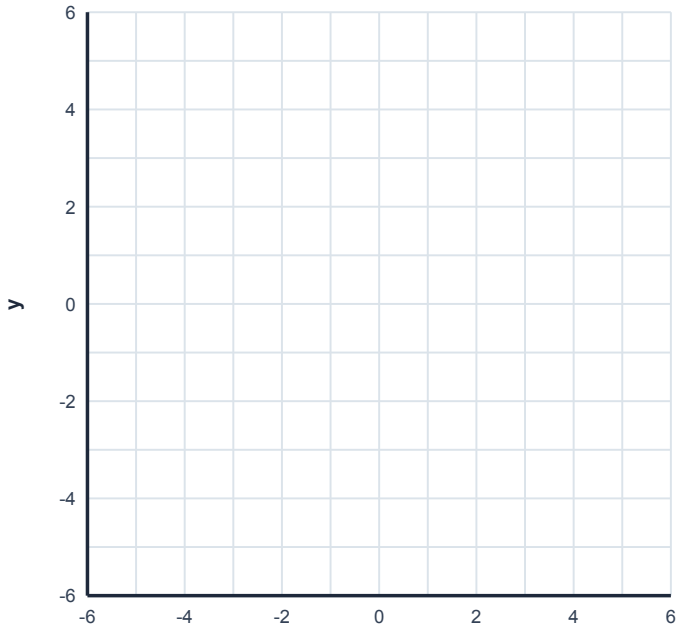
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PART 1 Understand the idea

models and meaning

1 **Plot and label each point on the grid.**

Plot (1, 4) and its reflections: over the x-axis, over the y-axis, and over both axes



SHOW YOUR THINKING

2 **Complete the table. Show how you found each value.**

Find the reflection of each point over the given axis.

Original Point	Axis	Reflected Point
(3, -7)	x-axis	
(-5, 2)	y-axis	
(8, -1)	x-axis	
(-4, -6)	y-axis	
(0, 9)	x-axis	
(-3, 0)	y-axis	

SHOW YOUR WORK

3 **Write the letter of the matching item on each line.**

Match each reflection rule to its description.

_____ Over x-axis
_____ Over y-axis
_____ Over both axes
_____ Point on x-axis
_____ Point on y-axis
_____ (0, 0)

(A) y-coordinate changes sign
(B) x-coordinate changes sign
(C) Both coordinates change sign
(D) Reflects to itself over x-axis
(E) Reflects to itself over y-axis
(F) Reflects to itself over any axis

PART 2 Apply it

real-world problems

4 **Circle the letter of the best answer. Show how you know.**

Point A is at (-3, 5). It is reflected over the x-axis to A', then A' is reflected over the y-axis to A''. What are the coordinates of A''?

(A) (-3, -5)
(B) (-3, 5)
(C) (3, -5)
(D) (3, 5)

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

5 **Find the mistake. Explain it, then show the correct work.**

Spot the Common Mistake

1. **Problem** Reflect the point (-3, -8) over the x-axis.
2. **Attempt** Both coordinates are negative, so I will flip the x-coordinate: -3 becomes 3. The answer is (3, -8).
3. **Check** Folding over the horizontal x-axis should send the point up across the axis, but this answer kept it below the axis.

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

What pattern do you notice about the coordinates when reflecting over the x-axis versus the y-axis?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 7.9

Reflect Points Across Axes

Challenge · Extension

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can reflect points across the x-axis and y-axis on the coordinate plane.

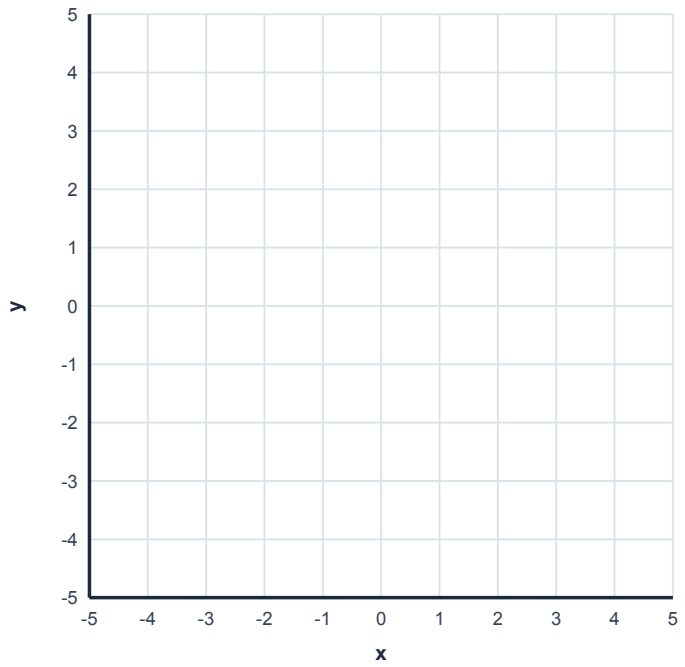
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PART 1 Understand the idea

models and meaning

1 **Plot and label each point on the grid.**

Plot the point (-2, 3). Reflect it over the x-axis, then reflect that result over the y-axis. Where do you end up? Now start over and reflect (-2, 3) over the y-axis first, then over the x-axis. Do you get the same final point?



SHOW YOUR THINKING

PART 2 Explain and correct

reasoning

2 Find the mistake. Explain it, then show the correct work.

Find the Reflection Error

1. **Problem** Reflect (-3, 5) over the y-axis
2. **Student thinking** The y-axis is vertical, so I change the y-coordinate
3. **Student answer** (-3, -5)

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

3 Answer in complete sentences.

A triangle has vertices at A(1, 2), B(4, 2), and C(4, 6). If you reflect the entire triangle over the y-axis, what are the new vertices? What do you notice about the size and shape of the reflected triangle compared to the original?

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

What pattern do you notice about the coordinates when reflecting over the x-axis versus the y-axis?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help